

MATE 2025 REGIONAL ROV COMPETITION



TigerSharks ROV CO

Mamaroneck High School, Mamaroneck, New York, United States of America

Technical Report

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I. Abstract

We are the Mamaroneck TigerSharks ROV Company from Mamaroneck, New York. We work at Mamaroneck High School to develop new tools and additions to our ROV (Remotely Operated Vehicle) to be more helpful in completing tasks like grabbing objects and moving them around. For our ROV, company members worked together over the course of many weeks to design and fabricate an ROV that was light, maneuverable, and relatively inexpensive to manufacture. Once we finished our thorough design process, we methodically assembled our ROV and used rapid prototyping to iterate quickly. Our main focus this year was to expand on our previous designs and increase reliability, usability, and efficiency. We optimized our electrical systems, redesigned our command and control structure for increased robustness, and added multiple new tools, such as a water sample pump, to our ROV to increase potential workload possibilities. Our company also designed and built a non-rov device (NRD) to allow for autonomous monitoring and water state awareness.



II. Company Members

	Max Lazarus - Senior President Max has previously completed three years of computer science, is now in engineering, and played a role in kick-starting the MATE team at the high school once again. Max has also been a part of the FTC robotics club at MHS, mostly contributing to the coding aspect of the team. He is also a part of the Mamaroneck High School sailing team and enjoys spending time on the water.
	Mustafa Khan – Junior Vice President Mustafa is currently in his first year of computer science and played a key role in CAD and fabrication of parts. Mustafa is also part of the FTC robotics club, in which he mostly contributed to systems integration and CAD of the main robot. Outside of MATE, Mustafa is part of the MHS debate team and other computer science related initiatives.
	Miles Contopoulos – Senior Lead Software Officer Miles was previously in the computer science program, but is now in the Engineering program at MHS. He helped start the MATE team and is currently in charge of software integration and development. He enjoys coding in his free time and is a member of the varsity track team. During the winter season, Miles also programs on the MHS FTC robotics team.
	Alex Lin – Senior Co-Lead Engineering Officer Alex is a senior in the engineering program at Mamaroneck High School and worked on the CAD for the main structure. He enjoys woodworking in his free time and is also a member of the varsity track team. He has participated in FTC and other STEM programs at MHS.

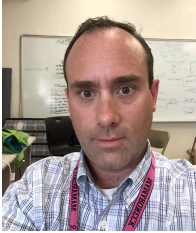


	Liam DeLaMora – Senior Co-Lead Engineering Officer Liam is in the engineering program at MHS, and worked on the CAD for the ROV and its attachments. He enjoys woodworking, tennis, and martial arts. Liam is part of the science research program at MHS, researching prosthetics and robotic rehabilitation.
	Max Prager - Junior Software Max is in the computer science program at MHS. He worked to design some aspects of the software and hardware of the MATE robot. He enjoys programming and designing in his free time outside of school. He also participates in other stem related programs at MHS, such as FTC robotics.
	Graham Welsh - Senior Engineering Graham is a sophomore at MHS who joined the MATE team when it was beginning to be reinstated. He is a major part of the FTC team who primarily builds robots. Outside of robotics he can be found teaching skiing at Windham Mountain or sailing on MHS' sailing team.
	Nate Metsch - Senior Engineering Nate is a senior at MHS who is in the engineering program. When he realized that the MATE team was returning, he wanted to help. He helped print some of the parts for the ROV and painted the chassis. Outside of robotics he enjoys 3d printing and making things.
	Jordan Metsch - Freshman Engineering Jordan is a Freshman at MHS. He joined the MATE team after learning about it when competing in the FTC competition in the MHS Robotics Club. When not participating in robotics, he enjoys completing engineering projects and 3D printing.



	Ethan Bregman - Freshman Engineering Ethan is a Freshman at MHS, and the MATE ROV competition is his first robotics competition. Co-leading the design and production of the peristaltic pump attachment with Jordan Metsch, he has learned a lot in the past few months as part of Mamaroneck Robotics. In his downtime, he runs the American Rocketry Challenge Club as co-president alongside Jordan Metsch. Also, he enjoys building remote-controlled aircraft and running for Mamaroneck XC/TF.
	Hugh Scobie- Sophomore Engineering Hugh is a sophomore at MHS who joined MATE after competing in the FTC Robotics competition in MHS robotics Club. He helped with the ideation of the utilities during meetings. He enjoys breaking and building things and has a Huge interest in Robotics and Civil engineering. He also rock climbs in his free time.
	Luke Minchilli-Sophomore Engineering Luke is a sophomore at MHS who joined MATE after competing in the FTC robotics competition in the MHS robotics club. He worked to design utilities for the robot using Fusion 360. He enjoys designing and building things and has a big interest in robotics and aerospace engineering. He is also part of the varsity wrestling team at MHS.
	Ian Soto - Junior Documentation Ian is a junior at MHS who joined MATE after Mustafa Khan piqued his interest. He mainly helped out with documentation and other little tasks around the testing of the robot. He also is proficient in coding in Python and Java. Outside of school, he takes a special interest in Calculus, as well as going climbing occasionally.



	Chloe Glazer - Senior Outreach Chloe is a senior at MHS who is in her fourth year of computer science at MHS. Outside of documenting for the FTC and MATE teams, she enjoys writing for the Globe newspaper and participating in the science research program.
	James Love – Engineering Teacher Mentor Mr. Love has degrees in Environmental Engineering and Education. When not teaching, he enjoys breaking and building things. He also enjoys snowboarding, hiking and sailing.



III. Missions

Shipwrecks, Spotter Buoys, and Flying Fish: Documenting the Impact of Climate Change and Invasive Species on the Great Lakes

The company must identify an unknown shipwreck that will be placed on the bottom of the pool using their ROV. The shipwreck handbook will include 9 unknown shipwrecks; the shipwreck located on the bottom of the pool will match one of these shipwrecks. After identifying the shipwreck, the company must also identify the length, type, and cargo carried by the ship. The company must also produce a photosphere of the area surrounding the shipwreck. A Smart buoy will be located in the pool at the start of the product demonstration. A rope will connect the surface, mid-water, and bottom base of the Smart buoy. A new thermistor, constructed identically to the one located on the Smart Buoy in mid-water, will be on the surface, side of the pool. Companies must replace the damaged mid-water thermistor on the Smart buoy. The company must also install a new pco2 sensor within the buoy. Finally, the ROV must collect a sample of water from within a container, identifying pH of the water contained.

Marine Renewable Energy: Producing Power from Our Planet While Monitoring Environmental Impact

The company must connect a floating solar panel array to the grid. The floating solar array panel will be on the surface adjacent to an offshore wind turbine. The ROV must retrieve a power connector, remove a cover on it's receiving port, and plug it in to activate the solar panel array. Then, the offshore wind farm subsurface structure must be protected against corrosion by replacing a sacrificial anode and applying an epoxy patch to a corroded area. The company must also mitigate corrosion on the base structure of the offshore wind farm through identification of a corrosion instance and then repair with an epoxy patch. The company must also then research and monitor the environmental impact of these off shore power production installations. The ROV must collect life stages of a jellyfish, collect fish species, and place a hydrophone to monitor the environmental impact of offshore energy structures. Then, the ROV must collect polyp stage jellies attached to the bottom of a solar panel array.

MATE Floats!

The company must construct a vertical profiling float, which communicates with the mission station prior to descending. Then, the float must complete two vertical profiles, with or without a buoyancy engine, with each repetition communicating the time to the mission station. The company must also submit documentation on the use, safety, and goals of the float.



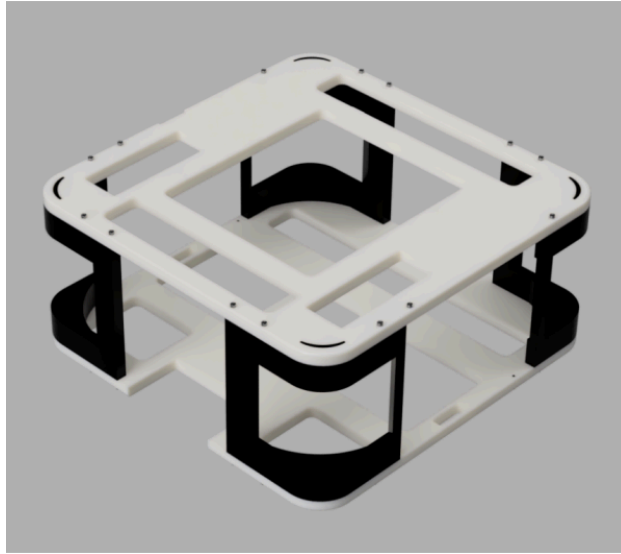
IV. Teamwork

Being a school team, the time that our company had to work on our ROV is a precious and limited resource. At the beginning of our season, our company created a Microsoft Teams workspace to help plan out deadlines, manage our schedule, and effectively communicate with each other. This workspace features everything from when individual designs were due to be completed, to the date we ordered specific parts. Utilizing this Teams workspace also allowed us to efficiently manage our time and delegate tasks to specific members. Our company also utilized many online tools to efficiently communicate and collaborate. These tools include the previously mentioned Microsoft Teams as well as the Google Drive suite to write our technical documents, specification sheet, and to design our posters collaboratively. For design and development, we use GitHub and Fusion360 extensively to ensure that all of our team members stay up to date with technical specifications and developmental changes. As with any project where multiple people are working together on one final product, our company had many differing views on topics as large as the overall design of the ROV to the smallest details. While sometimes these differing views lead us to creating unique solutions to problems that arise throughout the season, they can also lead to problems with integrating parts of the ROV together as a final product. Our company overcame this problem by keeping in constant communication with each other as problems arise, and by making sure that our engineers have relatively similar views on how we design individual components and how they link together.



V. Design Rationale

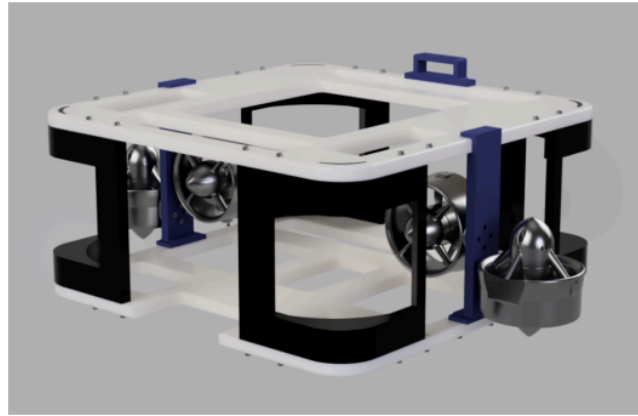
Structure



The main structure was a rigid frame fabricated with Delrin and Tough PLA. These materials were selected after extensive research based on density, weight, machinability, and price. The top and bottom plates were designed and machined on a Forest Scientific CNC router with Delrin. The connecting panels were 3D printed with Tough PLA, and attached to the Delrin via threaded inserts. The corner panels were designed to reduce turbulent flow in the water and minimize mass.



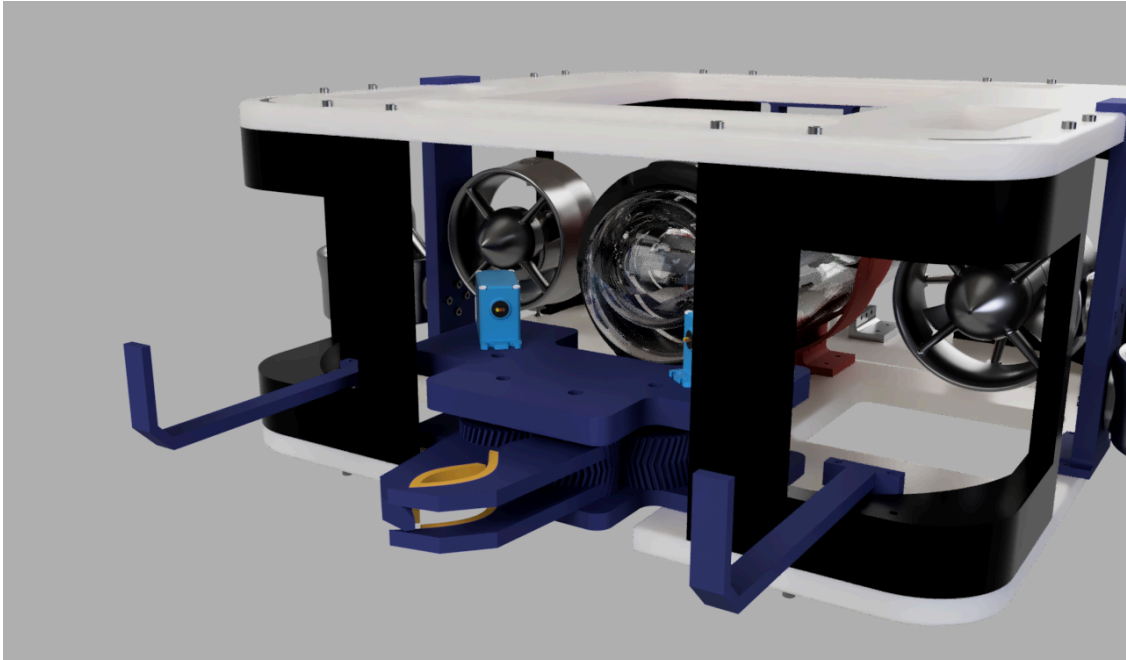
Propulsion and Buoyancy



The thruster layout was a big factor during the initial design stage. At first, engineers were focusing on mounting four thrusters in the corners, all rotated 45° from the center lines, to allow for omnidirectional movement. Later in the design process, the company settled on a more stable layout, having two facing up and two facing the front; this layout was chosen for its flexibility, balance, and ease of control. The layout is such that the downward vector of the upward-facing thruster cannot interfere with the front-facing thruster, an important factor for ensuring reliable movement. Also, the upward thruster locations being within the frame, pointing towards cutouts in the delrin, allows for more efficient use of space and allows the center of thrust to be as close to the center of mass as possible. The thruster of choice for our company was the Blue Robotics T200, which was picked for its high thrust-to-weight ratio. To comply with safety regulations, we soft-lock the motors into a thrust range where they can only operate with a maximum of 8 amps. When designing for buoyancy, the company engineers decided on an ROV that is slightly positively buoyant. This would act as a safety measure and ensure the ROV would float to the surface in the case of any onboard failure.



Hook/Claw

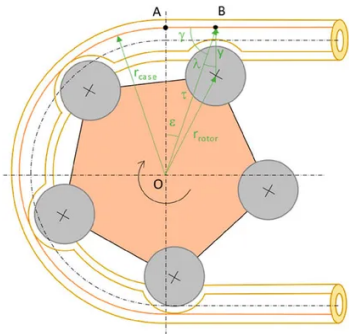


This season, as many missions require a claw, the company's focus was to make the claw as robust as possible. The claw is 3D printed with PLA filament, due to its rigidity, precision, and ease of printing. Double helical gears were chosen to reduce backlash, prevent gear slippage, increase the strength of the gear system without introducing uneven vertical forces (such as standard helical gears). The most important feature that ensures the claw is as strong as possible is the placement of the servos on auxiliary gears, away from the main gears of the claw gripper. Not only do these gears allow the claw to have more torque (2 : 1 gear ratio), but they also shield the servos from radial or axial loads, ensuring that the spline or shaft of the servos is not damaged. The gears attached to the gripper and their housing absorbs the force from any accidents, impacts, or motion, rather than the spline of the servo. The company also carefully optimized the design of the claw jaws to be as effective as possible. The claw has been designed to grab and pick up PVC pipe shaped between $\frac{1}{2}$ in and 2 in, as well as loops of wire or plastic attached to other objects. For this reason, an interlocking claw design was chosen, where the jaws of the claw overlap in the horizontal plane, so that no small objects can fall out of the claw's grip. Towards the center of the claw, the claw grip widens and curves, such that a variety of tubes of different diameters can be effectively grasped. For additional grip strength, a compressible interface was added between the curved section of the claw and objects to be picked up, made of flexible TPU filament. To ensure that the jaws of the claw always stayed aligned and the servos

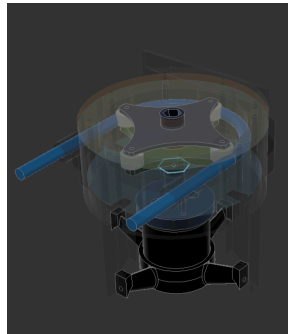


could share any torque applied, the two grippers on the claw are geared to each other as well as their respective servo. This distributes torque from the servos and prevents any issues with misalignment during the product demonstration. Finally, to maximize the torque of the claw, two servos were used, alongside a 2 : 1 gear ratio between the servos and the gripper.

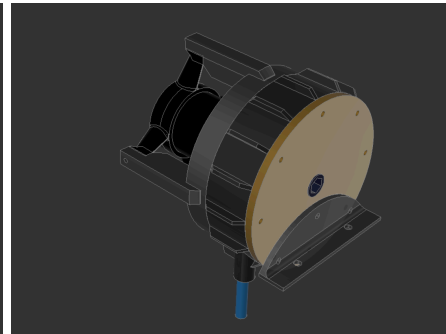
Pump



General Peristaltic Pump Diagram -Frontier



TigerSharks Pump (Case, Cover and ROV mount 20% Opacity, hex shaft hidden)

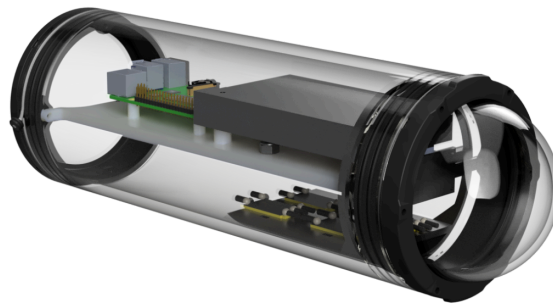


Full diagram of TigerSharks Peristaltic Pump (Hex shaft hidden)

This form of pump is known as a peristaltic pump, and it works on a slightly different principle than standard pumps. The peristaltic pump features a flexible hose on its interior, which is squeezed by a series of rollers to generate pressure. This setup, commonly used in the medical field, allows for a contamination-free stream which is necessary for an accurate chemical reading of our sample. In our pump, these rollers are two nylon-glass ball bearings stacked on top of one another. We chose nylon-glass as they have low rolling resistance while still being reasonably temperature resistant and being waterproof. We decided on 4 spokes which reduces resistance as our motor has a very high rpm which makes adding more rollers unnecessary. The 4 spoke roller carrier is spun via a hex shaft running through its center connected to a Blue Robotics T-100 thruster motor.



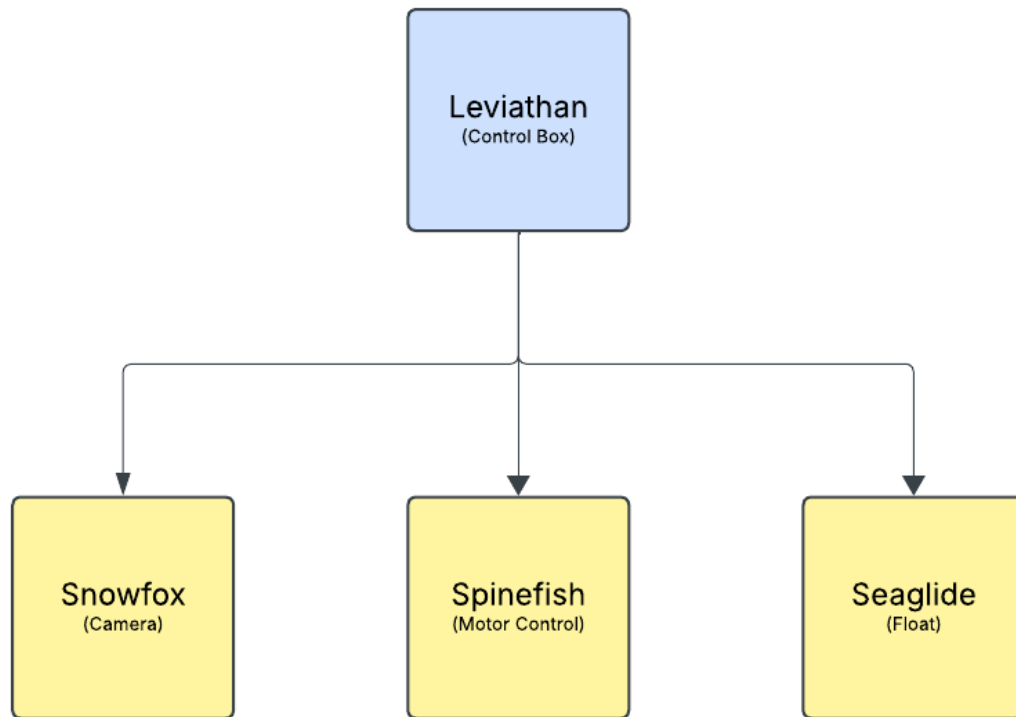
Electronics



The ROV has an acrylic enclosure that holds the Raspberry Pi, cameras, ESCs (Electronic Speed Controllers), a buck converter, and an Arduino. The wires from the thrusters are fed into the tube and wired into the ESCs, which output the proper 3-phase power required. The ESC uses 12V for power and PWM (Pulse-Width Modulation) signals to control the speed of the thrusters. The Arduino communicates with a servo board, which is used to output PWM to the thrusters, and the ESCs allow for easy and rapid control of motor speed. Both the Arduino and RPI take 5V power directly from a buck converter, which takes a parallel conductor input, split over multiple power lines joined at the ends, to limit voltage drop over the cables and allow for a stable power supply for the computers. For our tether system, the ROV utilizes five tethers: three sending power from our battery, one sending Ethernet data, and an RS-485 cable. The ROV features an onboard strain relief hub to prevent damage to the cables and structure during operation. One of the power cables and the Ethernet tether were reused from previous years, while the RS485 cable and the remaining power cables were added this year. As an additional measure to guarantee stable and repeatable movements from the ROV, buoyancy foam was zip-tied to the tethers to prevent them from interfering with or weighing down the ROV by making the cable neutrally buoyant. Finally, there is a 3D-printed camera mount that is mounted onto the dome and is the primary camera for the robot.



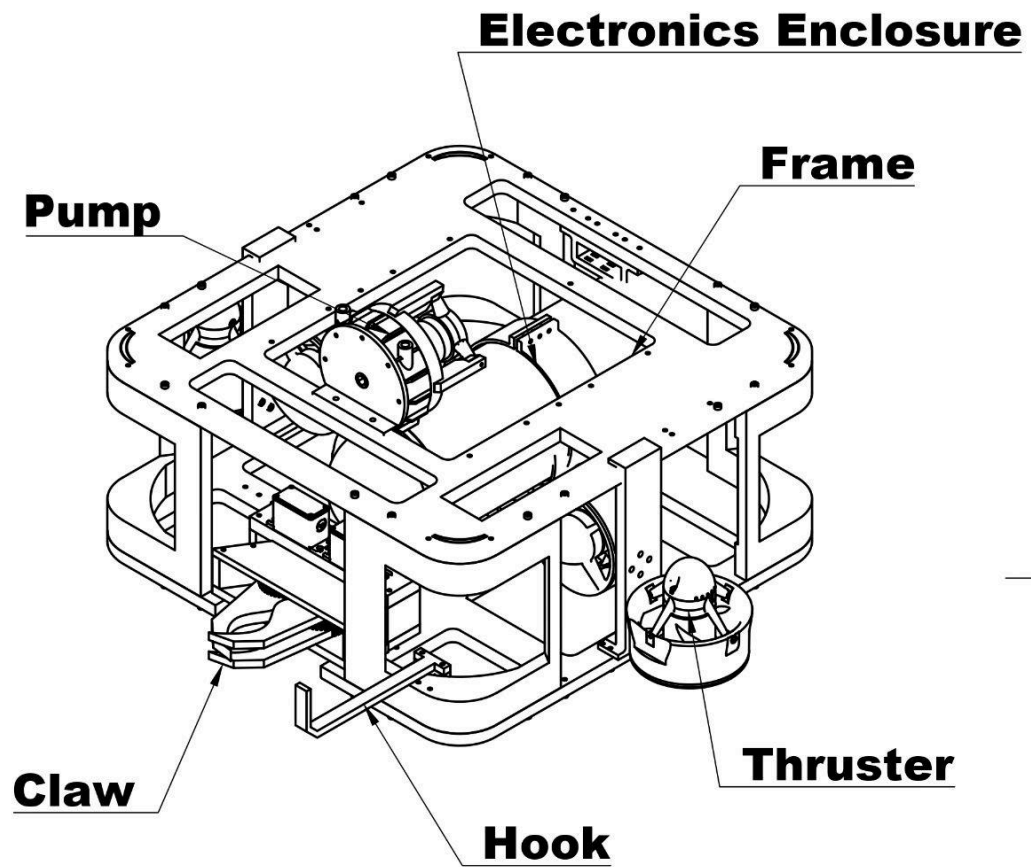
Control System



For the software and control system, our software engineers developed multiple custom libraries called Leviathan, Seaglide, Snowfox, and Spinefish. The libraries all work in tandem with each other, with Leviathan, which is run on an RPI5, driving all of them and being the primary system in the control box on the surface. Seaglide, which is run on an Arduino, is our custom driving library for the float, which communicates via radio to Leviathan when the float is on the surface of the water. For the bot, both Snowfox and Spinefish ensure the robot works as planned and have separate systems for optimal efficiency. Snowfox, which is run on an RPI5, deals with both cameras and sending their data over the Ethernet cable to Leviathan. Finally, Spinefish is run on an Arduino and deals with motor and servo control based on the data received over the RS485 cable.



Technical Drawing





VI. Accounting/Materials

Before starting the design and fabrication process, company members examined the ROV from the past year and recycled all components that were salvageable, such as the tethers and ESCs, thrusters, and the frame.

Material Expenses:

Item	Qty	Cost/Unit	Total Cost
Stepper	1	\$12.50	\$12.50
8mmx5mm Coupling	1	\$6.99	\$6.99
Lead Screw	1	\$11.87	\$11.87
Pressure Sensor	1	\$29.95	\$29.95
Tubing	1	\$4.49	\$4.49
PVC Pipe	1	\$12.81	\$12.81
Flex End Cap	2	\$9.81	\$19.62
Bearing	1	\$8.99	\$8.99
Fuse	1	\$8.99	\$8.99
8mm x 8mm Coupler	1	\$8.99	\$8.99
Oring	1	\$7.99	\$7.99
NRF Breakout	1	\$5.99	\$5.99
NRF wire	1	\$6.99	\$6.99
Bearing	1	\$7.19	\$7.19
PVC Pipe	1	\$12.81	\$12.81
Flex End Cap	1	\$9.81	\$9.81
Camera	1	\$43.59	\$43.59
Motor Cables	1	\$7.99	\$7.99
SMA Cable	1	\$6.99	\$6.99
Lora Breakout	2	\$24.33	\$48.66



Item	Qty	Cost/Unit	Total Cost
SMA for LORA	1	\$10.99	\$10.99
Stepper	1	\$9.99	\$9.99
Tubing	1	\$4.49	\$4.49
Battery	1	\$19.99	\$19.99
Motor Driver	1	\$7.99	\$7.99
Hose clamp	1	\$5.99	\$5.99
Rotary Seal	1	\$7.49	\$7.49
8mm Rod	1	\$3.79	\$3.79
Batteries	1	\$9.99	\$9.99
Power Cable	1	\$17.99	\$17.99
Power Plug	1	\$9.40	\$9.40
RS485 Cable	1	\$34.65	\$34.65
RS485 Boards	1	\$8.99	\$8.99
14 Hole End Cap	1	\$16.00	\$16.00
ESC	1	\$38.00	\$38.00
Bullet Connectors	1	\$21.99	\$21.99
Wetlink Penetrator Seals	1	\$18.00	\$18.00
Blank Penetrators	3	\$6.00	\$18.00
Epoxy Penetrators	4	\$6.00	\$24.00
T200's	4	\$258.00	\$1032.00
Previous expenses		\$638.11	\$638.11
		Sub Total:	\$1946.85

Other Expenses:

Item	Total Cost
Bus	<i>Covered by School</i>



T-shirts		565.29
Competition Fee		500.00
	Sub Total:	1065.29
Total Expenses		3012.14



VII. Software

This season we expanded on the systems created during last-year's competition. As it is currently only the third ever year after our founding as a company, a lot of the software and systems we use are still in active development. Last year's control application, Leviathan, was a major upgrade from our first year library, Seamoith, which controlled all parts of the robot at once. Leviathan was built as a robust but singular system, something our company learned we required after the modular but overall un-reliable Seamoith system. Leviathan sits in the control box and serves as a central hub and aggregator for all data collected during the round. Leviathan was largely unchanged this year, but we did develop three new applications; Seaglide, Snowfox, and Spinefish, to serve as the replacements for the remaining uses of Seamoith, our previous homegrown all-in-one software suite, on the robot and auxiliary systems. All three of these new applications are designed to be standalone and robust, specialized in controlling one part of the system, as compared to the “one software, one system” approach of Seamoith. This separated our control flow into individual applications that could be more easily adapted to the environment, such as Arduino code for hardened control systems, Rust code for the efficient and fail-proof but higher-level imaging system, and C++ for the fast all-around control required by the ground station. Overall, with the final removal of the last remaining parts of Seamoith this year, along with the new control flows of the new system, we have seen this year's software the most robust and error resistant of all of the previous years. So far this year we have seen a complete elimination of unexpected loss of ROV control, a 300% increase in UI framerate, and a 20% increase in camera view frame rate, as well as a significant reduction in network latency and increase in available bandwidth.



VIII. Constraints

When constructing the ROV we ran into several constraints. Two of the main ones being our lack of time and lack of space. Since we couldn't find a collective space to meet, we had to meet on and off at each other's houses, assembling the ROV and finding time to do all of this in between our busy individual schedules. To compensate for the little amount of time that we had we created a Microsoft Teams timeline with when each step of the process should be completed and how to adjust the schedule if something didn't go exactly to plan. A challenge that has repeatedly posed a problem throughout the past few years of our team has been that, our company systems design rationale has often led to instances where we went over our allowed maximum amperage pull. We have fixed these problems by implementing more efficient computing systems, and limiting our electrical pull through software solutions. Another primary challenge that we ran into making reliable electrical and communication systems. We solved this issue by adding 3 sets of new cables, two of which are power conductors acting as parallel conductors to reduce voltage drop. The third cable is an additional communication cable that uses the RS-485 protocol to communicate between devices reliably over long distances. Additionally, we ran into many issues for our 3D printed parts such as the pump and the claw when it comes to the durability of these parts. Finally, we ran into many problems when creating the Float, such as how to make a buoyancy engine, as this was a new challenge that we had not tried since the mission was first released a couple of years ago. The buoyancy engine promoted problems due to the company needing to figure out how to make a buoyancy that was both strong enough and waterproof. This problem was ultimately solved through the use of separate chambers, a wet chamber and a dry chamber, and also using a rotary seal to connect the two.



IX. Critical Analysis

As with any ROV or product, issues are bound to arise during the development process, but they can be minimized. One way we decreased the likelihood of the problems is through our prototyping methods. Many of the parts are 3D printed, specifically those that connect externally to the frame. The most significant advantage of 3D printing is its speed. All of the 3D-printed components on the ROV went through multiple iterations, testing different parameters until maximum efficiency was achieved. For testing the ROV, we used a multi-stage process. Testing the buoyancy first allowed us to gradually modify the buoyancy of the ROV using foam to achieve neutrality. Later in the testing process, the ROV was placed in a pool with all its electronics, allowing our engineers to test all the thrusters. The new software control systems work in harmony to minimize the effects of errors and unexpected fluctuations. Our system design allows each system to specialize in one aspect of the ROV's task, increasing reliability and robustness. Through careful procedures, design, and implementation, the company was able to minimize errors.



X. Challenges

Institutional Knowledge:

Institutional knowledge is something that we constantly struggle with. While our school technically has had a MATE company in the past, it was canceled before Covid-19, and as such, no one on our team had ever participated in MATE before our founding in 2022. As such, almost all of our systems have been built from the ground up using knowledge that we had from before MATE, along with knowledge that we've gained in the last two years. While admittedly, we have learned a lot in the last two years about how to manage and run a MATE team, as well as build ROV's, we still struggle with issues relating to a lack of knowledge relatively often. One such example is the Software suite we use, as outlined above, which has been made by scratch by our company members and is the culmination of everything we have learned in the last few years.

Future recruitment and potential loss of knowledge:

We anticipate difficulties in the future with our knowledge base. As shown in the members section, most of our company members are currently seniors in high school, and as such will not be here next year. We expect that this could follow with many of our non-senior members being left without knowledge bases for how to work with the software, design the robot, and test the systems. To hopefully mitigate these problems, we have begun working on comprehensive documentation of software systems, as well as encouraging non-senior members to ask as many questions as possible. Our company leadership fully expects this loss of knowledge to be a problem in the future but we are trying our best to secure against it, and set up pathways in the future to minimize the effects of these problems.

Prioritization of digital control systems over analog:

Our company has faced problems due to our prioritisation of a purely digital approach over an analog approach to our communication suite. The digital approach has allowed greater iteration potential, and is theoretically more powerful within a data collection perspective. However, the prioritization of digital systems has led to multiple instances of bandwidth and hardware limitations due to the difficulty of transmitting multiple different types of data such as live camera feeds, controller inputs, and diagnostics. Throughout the past few years our company developers have prioritized their time in creating more efficient software solutions, as outlined in the software subsection above. While our control suite is still not perfect, we are continuously iterating and improving it, to make it hopefully more powerful and adaptable than a traditional analog system.



XI. Safety

Safety Features:

The safety features utilized by our ROV include netting draped over the thrusters, fuses, an emergency stop, and strain relief on the tether. The 3D printed guards prevent injuries to people and damage to property by creating a barrier between the thruster and its surroundings. The fuses prevent damage to the ROV by stopping the flow of current when it exceeds the amount of current that our ROV can draw at maximum (calculations below), indicating a short. The emergency stop allows us to stop all of the ROV's systems if control over the ROV is lost by cutting power. The strain relief prevents damage to the electrical connections by securing the tether to a point on the frame, such that any strain or tension on the cable is applied to the frame rather than delicate connections.

Safety Checklist

Pre-round checklist:

- ☐ Ensure tether safety
 - ☐ Ensure all connectors are secure and tightened
 - ☐ Ensure that tether does not have knots
 - ☐ Ensure that tether is neatly coiled and easily extended/retracted
 - ☐ Ensure that all tether bindings and floats are secure
 - ☐ Ensure that ethernet cord has no abrasions, breaks, or other continuity errors
 - ☐ Ensure that power cord has no abrasions, breaks, or other continuity errors
 - ☐ Ensure that RS485 cord has no abrasions, breaks, or other continuity errors
- ☐ Ensure team safety
 - ☐ Ensure that all team members on the pool deck are wearing appropriate attire
 - ☐ Ensure that all team members on the pool deck are wearing safety glasses
 - ☐ Ensure that all team members on the pool deck are aware of their surroundings
- ☐ Ensure ROV safety
 - ☐ Ensure that the power connector is connected securely to the battery and ground station



- ☐ Ensure that fuse is installed, within 30cm of the ground station, and operational
- ☐ Ensure that power switch works and is operational in emergency situations
- ☐ Ensure that control systems are not in boot mode
- ☐ Ensure that actuators and motors are not in boot mode
- ☐ Initiate ROV/ground station software
 - ☐ Ensure that all control software is running nominally
 - ☐ Ensure that controllers are not still in boot configuration
 - ☐ Ensure that the ground station and the ROV have a stable connection through the ground station software
 - ☐ Ensure that the camera is receiving a live video feed

During round checklist:

Software failure on ROV or ground station

- ☐ Ensure that there is a software failure and not a momentary loss of connection
 - ☐ Use SSH to force stop ROV software
 - ☐ Force stop ground station software
 - ☐ Restart ground station and ROV software
 - ☐ Make sure that company engineers are aware of the error or problem so that further diagnosis can be applied after the round

Failure of ground station or ROV hardware

- ☐ Hit the E-Stop immediately after crash
- ☐ Restart the ROV or ground station hardware
- ☐ Restart ground station and ROV software
- ☐ Make sure that company engineers are aware of the error or problem so that further diagnosis can be applied after the round

Loss of control of the robot

- ☐ Hit the E-Stop **immediately**
- ☐ Check system logs for the ROV and ground stations to try to determine the exact cause of loss of control
- ☐ Follow the steps outlined in **Failure of ground station or ROV hardware** or **Software failure on ROV or ground station** to reset and fix what was diagnosed in the previous steps



Loss of video feed

- ☐ Put down controller and other movement sensitive equipment **immediately**
- ☐ Check system logs for the ROV and ground stations to try to determine the exact cause of loss of video feed
- ☐ Follow the steps outlined in **Failure of ground station or ROV hardware or Software failure on ROV or ground station** to reset and fix what was diagnosed in the previous steps
- ☐ Make sure that appropriate company engineers are aware of the error or problem so that further diagnosis can be applied after the round

Post round checklist:

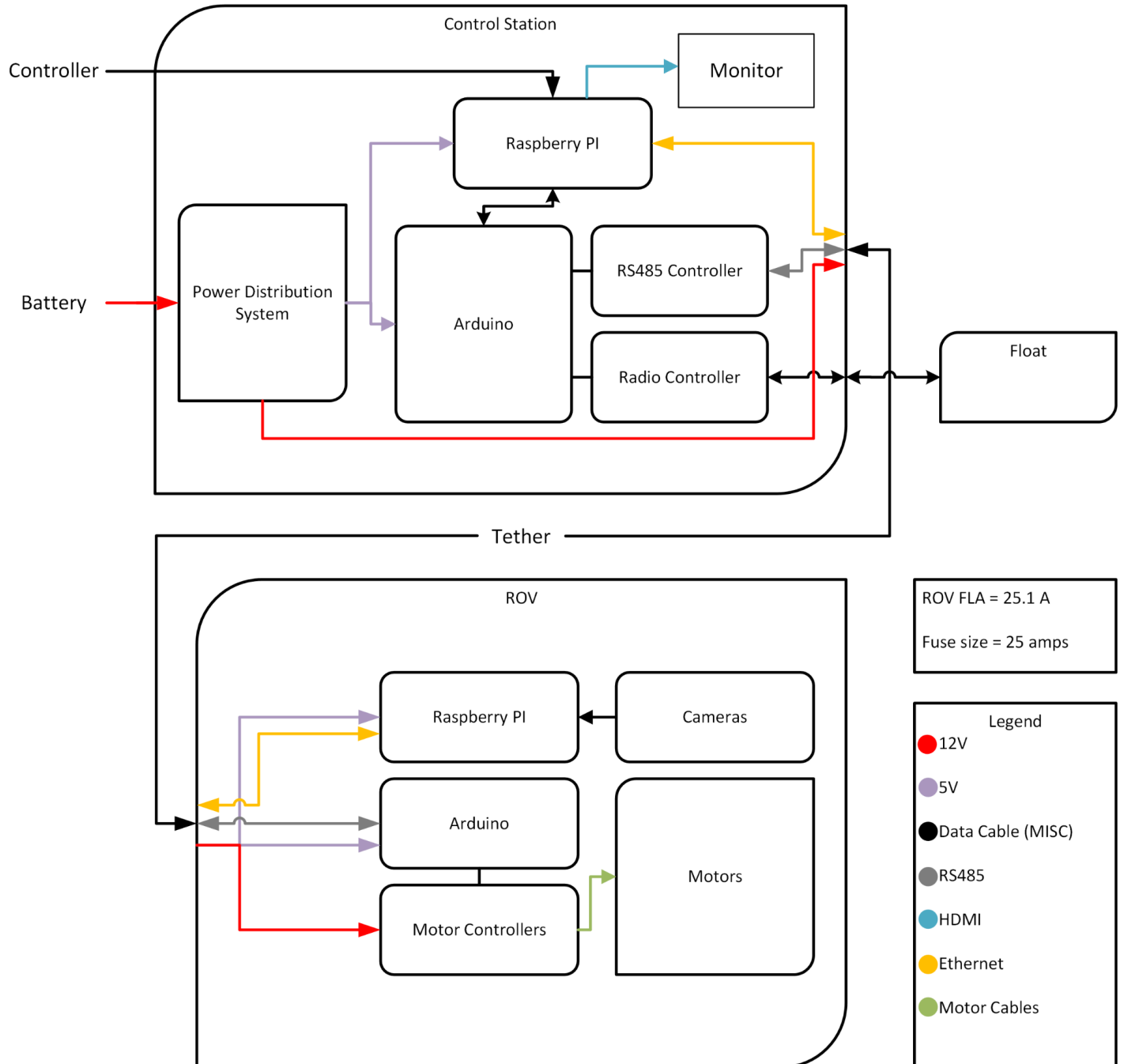
- ☐ Unscrew pressure relief valve to let latent air escape or enter
- ☐ Carefully remove control hardware from watertight tube
- ☐ Inspect for water intrusion
- ☐ Run a team meeting to diagnose and address problems encountered during round to ensure further safety during following rounds and events

Jobsite Safety and Environmental Analysis (JSEA):

We have developed a Jobsite Safety and Environmental Analysis outlining potential hazards during all times of interaction with company systems. It outlines safety measures for Testing, Bot Operations, Assembly, and Design/Development. Our goal within the JSEA was not only to ensure that all company members were operating within safe physical environments, but also that all personnel remained alert, free of fatigue, and not burnt out. It also establishes a flow and delegation of control, assigning and ensuring responsibilities with all company members. It was used during all times outlined on the JSEA, and a copy is included with this technical document.



XII. Systems Integration Diagram (SID)





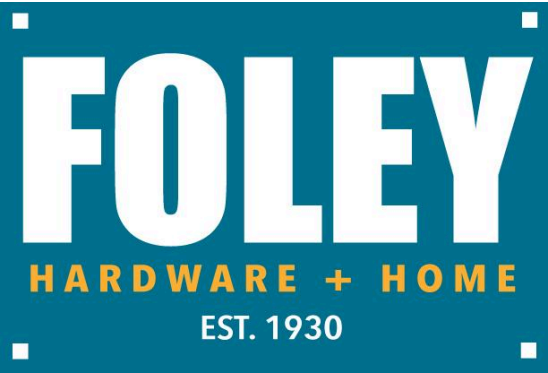
XIII. Fuse calculations

Item	Quantity	Individual Current @ 12V	Total Current @ 12V
Thruster	4	17A	68A
Raspberry Pi	2	5A	10A
Servo	2	2.2A	4.4A
Motor	1	11.5A	11.5A
Arduino	1	0.019A	0.019A
Screen	1	0.25A	0.25A
		Total:	94.169A
		Total x 150%:	141.2535A
		Fuse Size Used(limited)	25A



XIV. Sponsors

Total Received: \$1200

	Foley's Hardware
	Decicco's and Sons
	Zoffness College Prep
	McMicheal's Yacht Yards and Brokers



	Mamaroneck Boats and Motors
MAKEINSPIRES 	Makeinspires
 Mamaroneck Schools Foundation	Mamaroneck Schools Foundation



Final ROV

